

Madison K. Boudreaux

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PROFESSIONAL SUMMARY

Software Engineering Specialist with 5+ years of experience delivering mission-critical software for the defense and aerospace industry. Currently develops and maintains Java- and C++-based simulation and training systems for U.S. Air Force reconnaissance platform operators, including datalink communications and Distributed Interactive Simulation (DIS) protocol tooling, within Red Hat Enterprise Linux environments. Proven contributor to cross-functional development efforts, supporting execution against aggressive program milestones and the adoption of modern software engineering practices, with a strong foundation in configuration management and end-to-end software delivery. Active U.S. Secret Clearance (Top Secret) as of October 24, 2023.

CORE TECHNICAL SKILLS

Languages: Java, Python, C++, SQL

Software Engineering: Object-Oriented Design, Design Patterns, Simulation & DIS Protocol

Configuration Management: Git, GitLab, GitLab CI, Atlassian, JIRA, GitKraken

Tools & Practices: Eclipse, Red Hat Enterprise Linux (RHEL), Unit & Regression Testing, Agile/Scrum

PROFESSIONAL EXPERIENCE

L3Harris

Software Engineering Specialist

Mar 2026 – Present

Software Engineer on a Simulation team building training systems for U.S. Air Force reconnaissance platform operators. Develops and maintains Java- and C++-based applications on Red Hat Enterprise Linux that create realistic mission simulations, manage datalink communications, and record, filter, and play back Distributed Interactive Simulation (DIS) protocol traffic for lab and training environments.

- Develop and maintain simulation software used to train operators on U.S. Air Force reconnaissance platforms.
- Contribute to SEGUE, the team's primary simulation application, which builds realistic training scenarios and interfaces with airborne processors in lab environments.
- Support Datalink Simulator, a highly multithreaded system that manages datalink communications over UDP and TCP to reduce load on the main simulation platform.
- Develop tools that record, filter, view, and play back Distributed Interactive Simulation (DIS) protocol data units (PDUs) used in simulation and test environments.
- Build and enhance applications primarily in Java, with supporting work in C++, on Red Hat Enterprise Linux.
- Partner with Systems Engineering to design, implement, test, and release new capabilities, bug fixes, and software improvements.
- Use GitLab, GitLab CI, Eclipse, and JIRA, along with unit testing, regression testing, and weekly release processes, to support reliable delivery and reduce technical debt.

Lockheed Martin Aeronautics — Fort Worth, TX

Software Engineering Senior (2025–2026) | Software Engineer (2022–2025) | Software Engineering Associate (Jun 2020–2022)

Jun 2020 – Feb 2026

- Designed, developed, and delivered air-vehicle-level Radar Cross Section (RCS) test and evaluation software supporting Signature integration efforts, emphasizing quality, performance, usability, and cost efficiency.
- Developed and deployed software used for Signature acceptance at domestic U.S. sites and international customer locations.
- Drove execution of project milestones by contributing to development planning, coordinating schedules, and ensuring timely delivery of capabilities.

- Owned the release management process for the team's primary product and its four integrated modules, including build creation, testing coordination, packaging, and distribution.
- Created and maintained the team's test schedules for each software release to ensure verification coverage and schedule adherence.
- Mentored junior engineers through technical guidance, onboarding support, and code reviews, with a focus on readability, maintainability, and functional correctness.
- Developed and enhanced data analysis tools, graphical user interfaces (GUIs), and automated report-generation frameworks, including slide creation for customer-facing deliverables.
- Applied established engineering standards, processes, procedures, and tools across the full software development life cycle.
- Served as a primary technical point of contact for users, addressing bug reports, gathering feature requests, and providing general technical support within the organization.

Department of Computer Science, Lamar University — Beaumont, TX

Lab Technician | Grader

Jun 2018 – May 2020

- Prepared and maintained computer laboratories, including software installation and configuration to support coursework.
- Provided tutoring assistance to students in computer science courses.
- Evaluated and graded programming assignments according to departmental rubrics and technical requirements.

Texas Workforce Commission – Camp Code for Girls, Lamar University — Beaumont, TX

Camp Counselor & Co-Instructor

Jul 2018 – Aug 2018

- Educated and mentored middle school students in foundational computing concepts.
- Designed and delivered lesson plans covering variables, Boolean logic, and sprite motion using Scratch.
- Fostered problem-solving, analytical thinking, and interest in computer science through hands-on learning activities.

Cardinal Communities, Lamar University — Beaumont, TX

Peer Mentor

Aug 2017 – Jan 2018

- Planned and led educational and social activities to support first-year student transition to campus life.
- Served as a tutor and mentor, connecting students with departmental resources and providing individualized academic support.

Document Management Professionals — Houston, TX

Legal Production Assistant

Jan 2010 – Jan 2013

- Managed confidential legal documents, including organization, scanning, labeling, and optical character recognition (OCR) processing.
- Maintained strict adherence to confidentiality and document-handling procedures.

EDUCATION

Lamar University — Beaumont, TX

Bachelor of Science in Computer Science — May 2020

Major: Computer Science | Minor: Math